

Supercritical Propulsion Systems with GEM-Based Dynamic Control

Abstract—Conventional propulsion systems rely on discrete phase transitions and separate subsystems for fuel storage, thermal management, and tank pressurization. These architectures introduce complexity, instability, and efficiency limitations. This work proposes a unified propulsion framework based on the use of supercritical propellants, eliminating phase interfaces and enabling continuous thermodynamic control.

A dynamic model is developed in which the propellant remains in a supercritical state throughout the system, allowing its density to be modulated as a function of pressure and temperature. This enables controlled bifurcation of flow between regenerative cooling and autogenous pressurization subsystems without phase transitions.

The system is modeled as a nonlinear coupled dynamical system, and a linearized stability analysis is performed around an operational point. It is shown that the system can be tuned to achieve critically damped behavior, eliminating oscillations while maintaining rapid response.

Additionally, a geometric interpretation is introduced through the GEM (Geometric Environment Mapping) framework, representing system dynamics as trajectories in pressure–temperature space. This approach provides an intuitive and operational method for stability analysis and control.

The results demonstrate that supercritical propulsion architectures are physically consistent and dynamically controllable, offering a novel pathway toward integrated and simplified propulsion systems.

I. INTRODUCTION

Modern propulsion systems are fundamentally constrained by the coexistence of liquid and gaseous phases within propellant management architectures. These phase transitions introduce instabilities such as cavitation, boiling, and sloshing, while also requiring additional subsystems for pressurization and thermal control.

Recent advances in high-performance engines have approached supercritical regimes within combustion chambers, yet storage and feed systems remain largely phase-dependent. This work explores a propulsion architecture in which the propellant is maintained in a supercritical state from the tank through the entire system.

By eliminating phase boundaries, the system replaces discrete thermodynamic transitions with continuous control over fluid properties. This enables the use of density as a dynamic control variable, allowing the redistribution of propellant flow between subsystems.

The objective of this study is to establish a physically consistent framework for such a system, evaluate its dynamic stability, and introduce a geometric interpretation for control and monitoring.

II. PHYSICAL MODEL

A. Supercritical Regime

A supercritical fluid exists above its critical pressure and temperature, where no distinction exists between liquid and gas phases. Its density can be expressed as:

$$\rho = f(P, T) \quad (1)$$

Near the critical point, the following properties hold:

$$\frac{\partial \rho}{\partial P} > 0, \quad \frac{\partial \rho}{\partial T} < 0 \quad (2)$$

These derivatives can become large, enabling strong sensitivity to control inputs.

B. Operational Point

A stable operating point is selected in the supercritical regime:

$$P_0 = 60 \text{ bar}, \quad T_0 = 200 \text{ K} \quad (3)$$

This point lies beyond the critical conditions while avoiding singular behavior.

III. DYNAMIC MODEL

A. Mass Conservation

$$V \frac{d\rho}{dt} = \dot{m}_{in} - \dot{m}_1 - \dot{m}_2 \quad (4)$$

B. Density Evolution

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial P} \frac{dP}{dt} + \frac{\partial \rho}{\partial T} \frac{dT}{dt} \quad (5)$$

Using numerical approximations:

$$\frac{\partial \rho}{\partial P} \approx 4, \quad \frac{\partial \rho}{\partial T} \approx -4 \quad (6)$$

C. Energy Balance

$$\rho V c_p \frac{dT}{dt} = \dot{Q}_{in} - \alpha \dot{m}_1 - \beta \dot{m}_2 \quad (7)$$

D. Flow Model

$$\dot{m}_i \sim \sqrt{\rho(P - P_i)} \quad (8)$$

IV. LINEAR STABILITY ANALYSIS

The system is linearized around the operating point:

$$\mathbf{x} = \begin{bmatrix} \delta P \\ \delta T \end{bmatrix} \quad (9)$$

$$\dot{\mathbf{x}} = A\mathbf{x} \quad (10)$$

A representative stability matrix is:

$$A = \begin{bmatrix} -1.0 & 0.2 \\ -0.2 & -1.0 \end{bmatrix} \quad (11)$$

The characteristic equation is:

$$\lambda^2 + 2\lambda + 0.96 = 0 \quad (12)$$

This yields real negative eigenvalues, indicating a near critically damped system.

V. SYSTEM BEHAVIOR

Three regimes are identified:

- Subamortiguado: oscilaciones
- Sobreamortiguado: respuesta lenta
- Críticamente amortiguado: respuesta óptima

The system can be tuned toward the critically damped regime by balancing:

- dissipative terms (flow resistance)
- thermodynamic coupling

VI. GEM REPRESENTATION

The system can be represented in a geometric state space:

$$(P, T) \quad (13)$$

Where:

- density acts as a scalar field
- system evolution defines a vector field
- the operating point is a stable attractor

Trajectories in this space describe the dynamic response:

- direct convergence: stable
- spirals: oscillatory
- divergence: unstable

This representation enables intuitive monitoring and control of system behavior.

VII. DISCUSSION

The proposed architecture shifts system complexity from phase transitions to continuous thermodynamic control. While this increases control requirements, it offers several advantages:

- elimination of phase interfaces
- reduced mechanical complexity
- improved stability characteristics

The GEM framework provides an additional layer of interpretability, transforming system dynamics into a navigable geometric space.

VIII. CONCLUSION

This work demonstrates that propulsion systems based on supercritical fluids are physically viable and dynamically controllable. By eliminating phase transitions and using density as a control variable, the system achieves integrated functionality across subsystems.

The introduction of GEM provides a powerful tool for visualization and control, enabling new approaches to system design and operation.

Future work includes nonlinear simulation, real-fluid modeling, and experimental validation of subsystem behavior.